

EXPLORATORY DATA ANALYSIS

Dataset 2: Top 50 — 73 Countries (2023–2025)

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1. Overview

This document records the exploratory data analysis (EDA) performed on Dataset 2 — the Kaggle Top Spotify Songs in 73 Countries dataset — as part of the Hidden Gem Music capstone project. EDA was conducted in Power BI using Power Query aggregations across the full cleaned dataset. Dataset 2 is approximately 500MB and 2.1 million rows, which is within Power BI's capacity for all aggregations run.

The purpose of this EDA is to document what was actually observed in the data, flag discrepancies from expected values, and inform ingestion script decisions and stored procedure logic.

2. Dataset Summary

Metric	Observed Value
Total rows (post-cleaning)	2,110,316
Earliest snapshot_date	October 18, 2023
Latest snapshot_date	June 11, 2025
Unique countries (distinct country values)	73
GLOBAL rows (NULL country, filled during cleaning)	28,908
NULL daily_movement rows	0
NULL weekly_movement rows	0
Maximum credited artists on a single song	7 (artist_primary through artist_7)

3. Date Range

The earliest observed snapshot_date is October 18, 2023. The project reference documentation states October 17, 2023 as the dataset start date. This one-day discrepancy is likely a timezone difference in how Spotify published the first snapshot — chart data published late on October 17

in one timezone may have been recorded as October 18 in UTC. This is not a data quality issue and requires no corrective action.

The latest observed snapshot_date is June 11, 2025. This confirms the dataset coverage extends through mid-2025 as documented.

The 22-month data gap between Dataset 1 (ends December 31, 2021) and Dataset 2 (begins October 18, 2023) is confirmed. Time-series analysis spanning this gap is not possible and must be clearly marked in all visualizations.

4. Geographic Coverage

Category	Count	Notes
Distinct country values (ISO codes)	73	Includes GLOBAL
GLOBAL rows	28,908	Represent the Global Top 50 playlist — NULL country in raw data, filled with GLOBAL during Power BI cleaning. Map to NULL country_id in ChartEntry.
Country-specific rows	2,081,408	Total rows minus GLOBAL rows

Dataset 2 covers 73 distinct country values including GLOBAL. Dataset 1 covers 70 distinct region values. The 3-country difference is expected — the datasets have different geographic coverage and there is no requirement that they match. The Country lookup table is built from the instructor-provided CountryDataset.csv (245 countries) and covers all values observed in both datasets.

5. Movement Columns — Unexpected Finding

Both daily_movement and weekly_movement show zero NULL values across the entire 2,110,316-row dataset.

This is unexpected. The dataset documentation and project reference materials both state that these columns should be NULL on a track's first appearance in a country, because no prior day or prior week data exists to compute movement from. Zero NULLs means one of two things: either Spotify filled all first-appearance rows with the integer value 0 instead of leaving them NULL, or the dataset was pre-processed before being published to Kaggle in a way that backfilled these values.

This is a meaningful data quality finding. A movement value of 0 and a NULL movement value are not equivalent — 0 implies the song did not move, while NULL implies no prior data exists. If Spotify filled these with 0, then any stored procedure that uses daily_movement to detect first appearances (for example, as a Hidden Gem signal) cannot rely on NULL checks and would need a different approach to identify a track's first chart appearance in a country.

Recommended action: after BULK INSERT, run a query to identify rows where daily_movement = 0 and compare against the earliest snapshot_date per song per country. If they consistently

align, it confirms Spotify used 0 rather than NULL for first appearances. Document the finding and adjust any stored procedure logic that depends on NULL movement as a first-appearance signal.

6. Artist Column Distribution

The artists column was split into up to 7 positional columns during Power BI cleaning. The following counts show how many rows have a credited artist in each position beyond the primary.

Column	Rows with Value	% of Total Rows
artist_primary	2,110,316	100% — all rows
artist_2	858,692	40.7%
artist_3	250,660	11.9%
artist_4	76,867	3.6%
artist_5	37,578	1.8%
artist_6	15,757	0.7%
artist_7	7,249	0.3%

The maximum observed artist count is 7, compared to 4 in Dataset 1. This reflects a documented shift in music industry crediting conventions between the 2017–2021 and 2023–2025 periods. Multi-artist collaborations with 5 or more credited artists became significantly more common in Latin, trap, reggaeton, and hyperpop genres during the later period. This is not a data quality issue — it accurately reflects how music is credited in the current era.

7. Key Findings for Ingestion and Stored Procedure Authors

1. Zero NULL movement values — `daily_movement` and `weekly_movement` contain no NULLs. Ingestion scripts and stored procedures that rely on NULL movement as a first-appearance signal must be redesigned. Use `MIN(snapshot_date)` per song per country as the reliable first-appearance indicator instead.
2. 28,908 GLOBAL rows — these map to NULL `country_id` in `ChartEntry` and represent the Global Top 50 playlist. The schema accommodates this via the nullable `country_id` FK. Stored procedures that calculate per-country statistics must explicitly exclude or separately handle these rows.
3. Date range confirmed — October 18, 2023 through June 11, 2025. The one-day discrepancy from the documented start date of October 17 is a timezone artifact, not a data error.

4. 73 distinct countries including GLOBAL — all 73 values are covered by the Country lookup table. No FK violations are expected at ingestion for the country column.
5. Up to 7 credited artists per song — the ArtistSong join table and the 7 positional artist columns in the cleaned data are correctly aligned. The ingestion script must iterate through all 7 columns when populating Artist and ArtistSong rows.